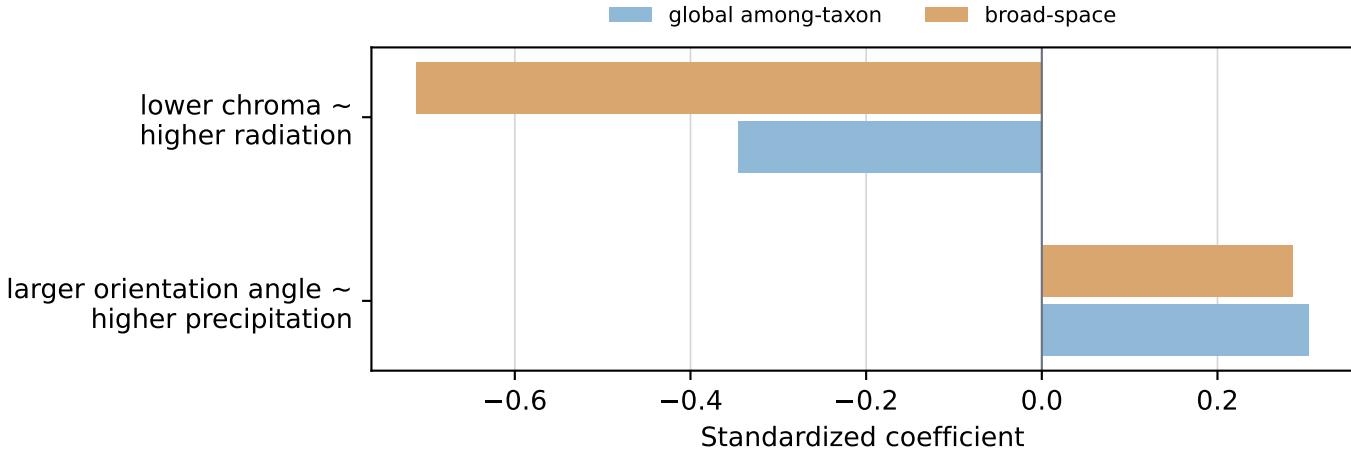


(a) Sequential attrition: why only two among-taxon rows remain



Sampling stability annotates rows; promotion requires broad-space and historical-placement gates.

(b) Direction after broad-space control



(c) Gate evidence retained in the frozen ledger

lower chroma ~ higher radiation

global $\beta = -0.345$; $q = 0.006$
sampling minimum ratio = 0.615; broad-space $\beta = -0.712$
spatial $P = 0.001$; residual Moran $P = 0.217$; trees = 52/52

larger orientation angle ~ higher precipitation

global $\beta = +0.304$; $q = 0.021$
sampling minimum ratio = 0.781; broad-space $\beta = +0.286$
spatial $P = 0.017$; residual Moran $P = 0.082$; trees = 52/52